



Designed for multi-signal strategy framework: momentum, trend strength, volatility, divergence.

Balanced between classic technicals and advanced Ehlers DSP indicators. At most 10 per category.

1 **Momentum (directional: long / short)**

Indicator	Type	Suggested role
RSI (14)	continuous / discrete	Overbought/oversold levels + cross-of-level
Stochastic (14,3,3)	continuous / discrete	%K / %D crossovers + level thresholds
CCI (20)	continuous / discrete	±100 threshold + zero-line cross
Williams %R (14)	continuous	−80 / −20 thresholds
Ehlers Fisher Transform (10)	continuous / discrete	Zero-line cross + overbought/oversold (±2)
Ehlers Cyber Cycle (20)	continuous	Zero-line cross, cycle phase
Ehlers Sinewave (20)	discrete	Lead/lag cross (I/E phase)
Ehlers Stochastic Cyber Cycle	discrete	Crossover of smoothed lines
Rate of Change (ROC, 12)	continuous	Sign change (zero-line cross)
MACD (12,26,9)	discrete	Signal line crossover (used as momentum)

2 **Trend Strength (regime only, direction = both)**

Indicator	Type	Role as filter
ADX (14)	continuous	High strength (>25) vs low strength (<20)
Aroon (25)	continuous	Strong trend (>70) vs weak (<30)
Ehlers Instantaneous Trendline (EIT)	continuous	Slope > 0 → uptrend regime, < 0 → downtrend regime
Ehlers TrendFlex	continuous	Strength measure (0-100) similar to ADX but zero-lag
Ehlers Market Mode Identifier	discrete	1 = trending, 0 = cycling – perfect regime filter
Super Trend (10,3)	continuous	Above/below band as trend regime
Choppiness Index (14)	continuous	Low (<38) = trending, high (>62) = choppy
Ehlers Band-Pass Filter (48,20)	continuous	Output > 0.5 = trending phase
Linear Regression Slope (20)	continuous	Slope > 0 for uptrend, < 0 for downtrend (regime)
ATR ratio (ATR / Close)	continuous	High ratio (>0.02) signals strong movement (non-directional)

3 **Volatility (regime only, direction = both)**

Indicator	Type	Role
Garman-Klass (20)	continuous	High/low regime (dynamic threshold)
Donchian Width (20)	continuous	High/low width regime
ATR (14)	continuous	High/low relative to moving average

Indicator	Type	Role
Bollinger Band Width (20,2)	continuous	Squeeze (low width) vs expansion (high width)
Ehlers SuperSmoother (10)	continuous	Smoothed volatility estimate (as filter)
Ehlers Roofing Filter (10,48)	continuous	High-frequency noise; high values = volatile
Parkinson's Historical Volatility (20)	continuous	Efficient range-based volatility
Yang-Zhang Volatility (20)	continuous	Robust to opening jumps
Ehlers Standardized Volatility	continuous	Z-score of volatility over 50 bars
Average Daily Range / 20-day MA	continuous	Regime based on ratio > 1.2 or < 0.8

4 Divergence (discrete, directional long/short)

Indicator	Type	Description
RSI Divergence	discrete	Conventional hidden / regular divergence
Stochastic Divergence	discrete	%K / %D divergence
CCI Divergence	discrete	Classic divergence (± 100 zones)
MACD Divergence	discrete	Histogram / line divergence
Ehlers Reflex	discrete	Built-in divergence detection (price – Reflex cross)
Ehlers Fisher Transform Divergence	discrete	Zero-lag divergence (more sensitive)
Ehlers Cyber Cycle Divergence	discrete	Cycle-based divergence

Indicator	Type	Description
OBV Divergence	discrete	On-Balance Volume divergence
Williams %R Divergence	discrete	Overbought/oversold zone divergence
Ehlers Sinewave Phase Divergence	discrete	Lead/sine wave phase mismatch

- All indicators can be implemented with the **@signal** decorator.
- Ehlers Market Mode + Ehlers Reflex are the highest-value additions for trend/cycle filtering and divergence detection.
- Save this HTML and print to PDF for offline reference.